

EE227 – Digital Logic Design

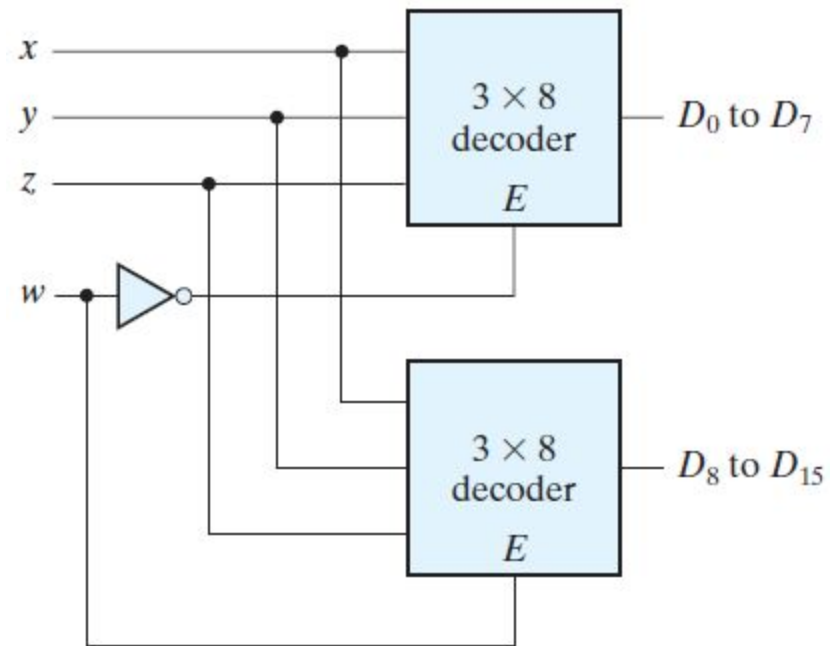
Lecture 16

Outline

- 4 to 16 line decoder
 - Combinational Logic Implementation with Decoders
 - Full Adder with Decoder
- Encoder
- Priority Encoder

4 to 16 Line Decoder

- Two 3-to-8-line decoders with enable inputs connected to form a 4-to-16-line decoder
- When $w = 0$, the top decoder is enabled and the other is disabled
- The bottom decoder outputs are all 0's, and the top eight outputs generate minterms 0000 to 0111
- When $w = 1$, the enable conditions are reversed, the bottom decoder outputs generate minterms 1000 to 1111, while the outputs of the top decoder are all 0's



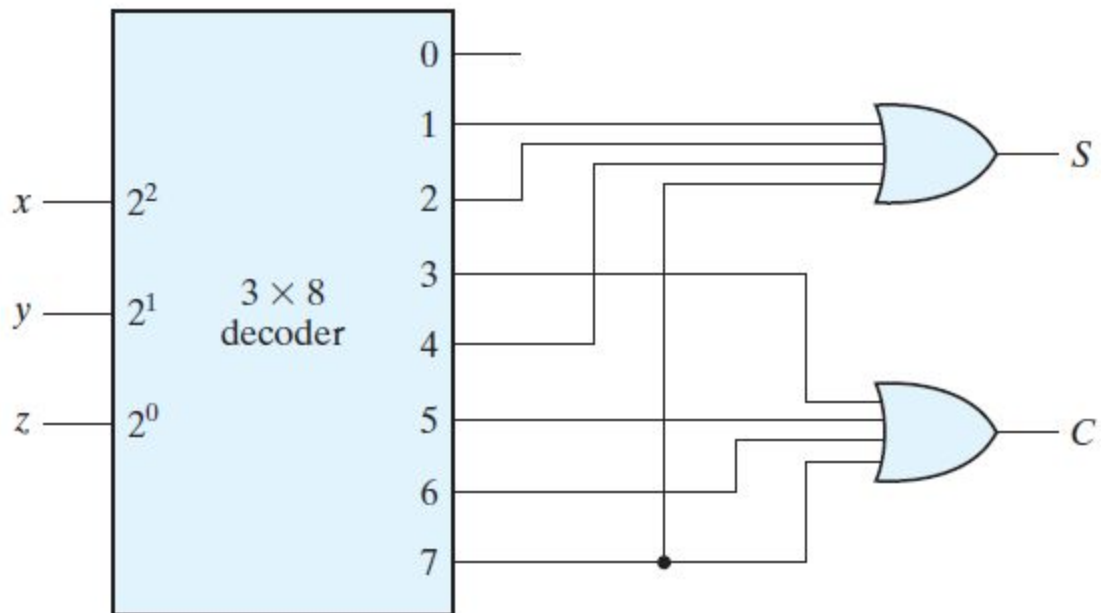
Combinational Logic Implementation with Decoders (1/2)

- A decoder provides the 2^n minterms of n input variables
- Each asserted output of the decoder is associated with a unique pattern of input bits
- Since any Boolean function can be expressed in sum-of-minterms form, a decoder that generates the minterms of the function, together with an external OR gate that forms their logical sum, provides a hardware implementation of the function
- In this way, any combinational circuit with n inputs and m outputs can be implemented with an n – to -2^n line decoder and m OR gates

Combinational Logic Implementation with Decoders (2/2)

- The procedure for implementing a combinational circuit by means of a decoder and OR gates requires that the Boolean function for the circuit be expressed as a sum of minterms
- A decoder is then chosen that generates all the minterms of the input variables
- The inputs to each OR gate are selected from the decoder outputs according to the list of minterms of each function

Full Adder with Decoder



x	y	z	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$S = \Sigma(1, 2, 4, 7)$$

$$C = \Sigma(3, 5, 6, 7)$$

Functions with Large Inputs

- A function with a long list of minterms requires an OR gate with a large number of inputs
- A function having a list of k minterms can be expressed in its complemented form F' with $2^n - k$ minterms
- If the number of minterms in the function is greater than $2^n/2$, then F' can be expressed with fewer minterms
- In such a case, it is advantageous to use a NOR gate to sum the minterms of F'
- The output of the NOR gate complements this sum and generates the normal output F

Implementation of Functions with Decoders with NAND Gates

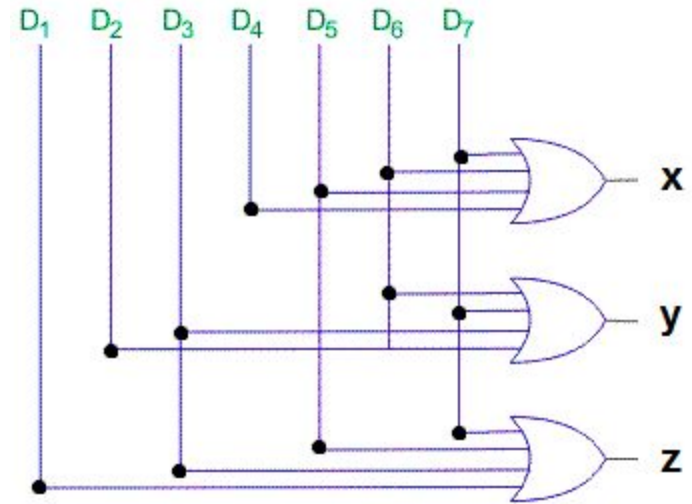
- If NAND gates are used for the decoder, then the external gates must be NAND gates instead of OR gates
- This is because a two-level NAND gate circuit implements a sum-of-minterms function and is equivalent to a two-level AND – OR circuit

Encoder

- An encoder is a digital circuit that performs the inverse operation of a decoder
- An encoder has 2^n (or fewer) input lines and n output lines
- The output lines, as an aggregate, generate the binary code corresponding to the input value

Octal to Binary Encoder

- $z = D_1 + D_3 + D_5 + D_7$
- $y = D_2 + D_3 + D_6 + D_7$
- $x = D_4 + D_5 + D_6 + D_7$



Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

Limitations of a Simple Encoder

- The encoder defined in last slide has the limitation that only one input can be active at any given time
- If two inputs are active simultaneously, the output produces an undefined combination
- For example, if D_3 and D_6 are 1 simultaneously, the output of the encoder will be 111 because all three outputs are equal to 1
- The output 111 does not represent either binary 3 or binary 6
- To resolve this ambiguity, encoder circuits must establish an input priority to ensure that only one input is encoded

Priority Encoder (1/4)

- A priority encoder is an encoder circuit that includes the priority function
- The operation of the priority encoder is such that if two or more inputs are equal to 1 at the same time, the input having the highest priority will take precedence
- Truth table of a 4 bit priority encoder is shown below
- V is a valid bit indicator
 - That is set to 1 when one or more inputs are equal to 1
 - If all inputs are 0, there is no valid input and V is equal to 0
- D_3 has the highest priority and D_0 has the lowest priority
- The priority is defined in this sequence D_3, D_2, D_1, D_0

Inputs				Outputs		
D_0	D_1	D_2	D_3	x	y	V
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

Priority Encoder (2/4)

Inputs			
D0	D1	D2	D3
0	0	0	0
1	0	0	0
X	1	0	0
X	X	1	0
X	X	X	1

Outputs		
x	y	V
X	X	0
0	0	1
0	1	1
1	0	1
1	1	1

=

Inputs			
D0	D1	D2	D3
0	0	0	0
1	0	0	0
0	1	0	0
1	1	0	0
0	0	1	0
1	0	1	0
0	1	1	0
1	1	1	0
0	0	0	1
1	0	0	1
0	1	0	1
1	1	0	1
0	0	1	1
1	0	1	1
0	1	1	1
1	1	1	1

Outputs		
x	y	V
X	X	0
0	0	1
0	1	1
0	1	1
1	0	1
1	0	1
1	0	1
1	0	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1

Priority Encoder (3/4)

D_2D_3		D_0D_1			
		00	01	11	10
m_0	00	X	1	1	1
	01	m_4	1	1	1
	11	m_{12}	1	1	1
	10	m_8	1	1	1

$$x = D_2 + D_3$$

D_2D_3		D_0D_1			
		00	01	11	10
m_0	00	X	1	1	
	01	m_4	1	1	m_6
	11	m_{12}	1	1	m_{14}
	10	m_8	1	1	m_{10}

$$y = D_3 + D_1D'_2$$

$$x = D_2 + D_3$$

$$y = D_3 + D_1D'_2$$

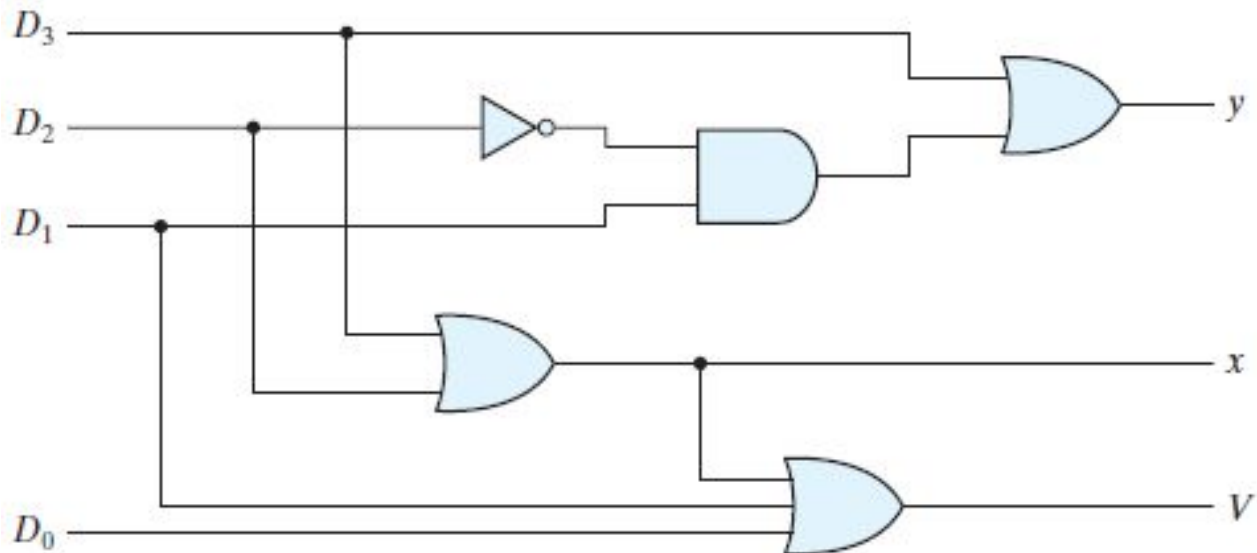
$$V = D_0 + D_1 + D_2 + D_3$$

Inputs			
D0	D1	D2	D3
0	0	0	0
1	0	0	0
0	1	0	0
1	1	0	0
0	0	1	0
1	0	1	0
0	1	1	0
1	1	1	0
0	0	0	1
1	0	0	1
0	1	0	1
1	1	0	1
0	0	1	1
1	0	1	1
0	1	1	1
1	1	1	1

Outputs		
x	y	V
X	X	0
0	0	1
0	1	1
0	1	1
1	0	1
1	0	1
1	0	1
1	0	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1

Priority Encoder (4/4)

- $x = D_2 + D_3$
- $y = D_3 + D_1 D'_2$
- $V = D_0 + D_1 + D_2 + D_3$



Multiplexer

- A multiplexer is a combinational circuit that selects binary information from one of many input lines and directs it to a single output line
- The selection of a particular input line is controlled by a set of selection lines
- Normally, there are 2^n input lines and n selection lines whose bit combinations determine which input is selected

2 to 1 Line Multiplexer

- A two-to-one-line multiplexer connects one of two 1-bit sources to a common destination
- The circuit has two data input lines, one output line, and one selection line S
- When $S = 0$, the upper AND gate is enabled and I_0 has a path to the output
- When $S = 1$, the lower AND gate is enabled and I_1 has a path to the output

